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Chip-on-film packages and display apparatuses including the same

Abstract

A chip-on-film package includes a base film having a top surface and a bottom surface, and a circuit region; a source driver chip and a gate driver chip mounted on the circuit region; a first conductive line on the top surface of the base film, a second conductive line on the bottom surface of the base film, and a conductive via that connects the first and second conductive lines to each other; a first row of bonding pads on the circuit region and connected to the source driver chip; a second row of bonding pads on the circuit region and connected to the source driver chip and the gate driver chip; and a test pad outside the circuit region and connected to the first and second conductive lines and the conductive via.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of and claims priority to U.S. patent application Ser. No. 16/874,120, filed May 14, 2020, which claims the benefit of Korean Patent Application No. 10-2019-0112367, filed on Sep. 10, 2019, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated herein in its entirety by reference.

BACKGROUND

(1) The present disclosure relates to a chip-on-film (COF) package and a display apparatus including the same, and more particularly, to a COF package in which different types of semiconductor chips are arranged and a display apparatus including the COF package.

(2) A chip-on-film (COF) package may include a semiconductor chip mounted on a base film, and the semiconductor chip may be electrically connected to an external apparatus through a conductive line and a conductive pad connected to the conductive line in the base film. In recent years, as bezel downscaling and panel thinning in display apparatuses have been increasingly demanded, the types and number of semiconductor chips mounted in one COF package have been gradually increased.

SUMMARY

(3) The inventive concept provides a new arrangement of conductive lines and conductive pads to ensure a predetermined bonding pitch in one chip-on-film (COF) package in which different types of semiconductor chips are arranged.

(4) The technical objectives of the inventive concept are not limited to the above and other objectives may become apparent to those of ordinary skill in the art based on the following descriptions.

(5) According to an aspect of the inventive concept, there is provided a COF package including a base film having a top surface and a bottom surface that are opposite to each other, and further having a circuit region. The COF package includes a source driver chip and a gate driver chip

mounted on the circuit region on the top surface of the base film, a first conductive line on the top surface of the base film, a second conductive line on the bottom surface of the base film, and a conductive via that connects the first and second conductive lines to each other. Moreover, the COF package includes a first row of bonding pads on the circuit region on the top surface of the base film and connected to the source driver chip, a second row of bonding pads on the circuit region on the top surface of the base film and connected to the source driver chip and the gate driver chip, and a test pad outside the circuit region on the top surface of the base film and connected to the first and second conductive lines and the conductive via.

(6) According to another aspect of the inventive concept, there is provided a COF package including a base film having a top surface and a bottom surface that are opposite to each other, the base film further having a circuit region. The COF package includes a source driver chip and a gate driver chip mounted on the circuit region on the top surface of the base film, a first conductive line on the top surface of the base film, a second conductive line on the bottom surface of the base film, and a conductive via that connects the first and second conductive lines to each other. Moreover, the COF package includes a first row of bonding pads on the circuit region on the bottom surface of the base film and connected to the gate driver chip, a second row of bonding pads on the circuit region on the bottom surface of the base film and connected to the source driver chip, and a test pad outside the circuit region of the bottom surface of the base film and connected to the first and second conductive lines and the conductive via.

(7) According to another aspect of the inventive concept, there is provided a display apparatus including a COF package having a base film, a display panel that faces a portion of a top surface of the base film, and a driver printed circuit board (PCB) that faces another portion of the top surface of the base film. The COF package includes the base film, a source driver chip and a gate driver chip mounted on the top surface of the base film, a first conductive line on the top surface of the base film, a second conductive line on a bottom surface of the base film, and a conductive via that connects the first and second conductive line. Moreover, the COF package includes a first row of bonding pads on the top surface of the base film and connected to the source driver chip, and a second row of bonding pads on the top surface of the base film and connected to the source driver chip and the gate driver chip.

(8) According to another aspect of the inventive concept, there is provided a display apparatus including a COF package having a base film, a display panel that faces a portion of a bottom surface of the base film, and a driver PCB that faces a portion of a top surface of the base film. The COF package includes the base film, a source driver chip and a gate driver chip mounted on the top surface of the base film, a first conductive line on the top surface of the base film, a second conductive line on the bottom surface of the base film, and a conductive via that connects the first and second conductive lines to each other. Moreover, the COF package includes a first row of bonding pads on the bottom surface of the base film and connected to the gate driver chip, and a second row of bonding pads on the bottom surface of the base film and connected to the source driver chip.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments of the inventive concept will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings in which:

(2) FIG. 1 is a schematic perspective view of a display apparatus including a chip-on-film (COF) package according to an embodiment;

(3) FIGS. 2A to 2E are schematic diagrams of a COF package according to an embodiment;

(4) FIG. 3 is a schematic plan view of a portion of a COF package according to an embodiment;

- (5) FIG. 4 is a schematic plan view of a COF package according to an embodiment;
- (6) FIGS. 5A to 5E are schematic diagrams of a COF package according to an embodiment;
- (7) FIG. 6 is a schematic plan view of a portion of a COF package according to an embodiment;
- (8) FIG. 7 is a schematic plan view of a COF package according to an embodiment;
- (9) FIG. 8 is a schematic plan view of a COF package according to an embodiment;
- (10) FIGS. 9A and 9B are schematic diagrams of a display apparatus including a COF package according to an embodiment; and
- (11) FIGS. 10A and 10B are schematic diagrams of a display apparatus including a COF package according to an embodiment.

DETAILED DESCRIPTION

- (12) Hereinafter, embodiments will be described in detail with reference to the accompanying drawings.
- (13) FIG. 1 is a schematic perspective view of a display apparatus **1000** including a chip-on-film (COF) package according to an embodiment.
- (14) Referring to FIG. 1, the display apparatus **1000** may include at least one COF package **100**, a driver printed circuit board (PCB) **400**, and a display panel **500**.
- (15) The COF package **100** may be a package including a semiconductor chip. A display chip may be a display driver IC (DDI). Different types of semiconductor chips may be arranged on the at least one COF package **100**. For example, the semiconductor chip may include a source driver chip and a gate driver chip.
- (16) The COF package **100** may be located between the driver PCB **400** and the display panel **500** and connected to each of the driver PCB **400** and the display panel **500**. The COF package **100** may receive a signal output by the driver PCB **400** and transmit the signal to the display panel **500**.
- (17) At least one driver circuit chip **410** may be mounted on the driver PCB **400**. The at least one driver circuit chip **410** may simultaneously or sequentially apply power and a signal to the COF package **100**.
- (18) The display panel **500** may be, for example, a liquid crystal display (LCD) panel, a light emitting diode (LED) panel, an organic LED (OLED) panel, and a plasma display panel (PDP).
- (19) The COF package **100** may be electrically connected to a driver interconnection line **430** of the driver PCB **400** and a panel interconnection line **530** of the display panel **500**.
- (20) In some embodiments, one COF package **100** may be connected between the driver PCB **400** and the display panel **500**. For example, when the display panel **500** provides a small-area screen for a small-sized device (e.g., a mobile phone) or supports a relatively low resolution, the display apparatus **1000** may include only one COF package **100**.
- (21) In other embodiments, a plurality of COF packages **100** may be connected between the driver PCB **400** and the display panel **500**. For example, when the display panel **500** provides a large-area screen for a large-sized device (e.g., a television) or supports a relatively high resolution, the display apparatus **1000** may include a plurality of COF packages **100**.
- (22) The COF package **100** may be connected only to one side of the display panel **500**. However, the inventive concept is not limited thereto, and in some embodiments, at least one COF package **100** may be connected to each of at least two sides of the display panel **500**.
- (23) The display panel **500** may include a transparent substrate **510** and an image region **520** and the panel interconnection line **530**, which are formed on the transparent substrate **510**. The transparent substrate **510** may be, for example, a glass substrate or a transparent flexible substrate. A plurality of pixels included in the image region **520** may be connected to a plurality of panel interconnection lines **530** corresponding thereto, and operate in response to signals provided by the semiconductor chip mounted on the COF package **100**.
- (24) An input pad may be formed at one end of the COF package **100**, and an output pad may be formed at another end thereof. The input pad and the output pad may be respectively connected to the driver interconnection line **430** of the driver PCB **400** and the panel interconnection line **530** of

the display panel **500** by an anisotropic conductive layer **600**.

(25) The anisotropic conductive layer **600** may include, for example, an anisotropic conductive film or an anisotropic conductive paste. The anisotropic conductive layer **600** may have a structure in which conductive particles are distributed in an insulating adhesive layer. Also, the anisotropic conductive layer **600** may have anisotropic electrical characteristics so that the anisotropic conductive layer **600** may be insulated in a first direction (e.g., an X direction (FIG. 2A)) between adjacent electrodes while enabling the formation of an electrical conduction path only in a second direction (e.g., a Z direction (FIG. 2A)) of electrodes during the connection. When an adhesive is melted by applying heat and pressure to the anisotropic conductive layer **600**, the conductive particles may be arranged between opposite electrodes, for example, between the input pad and the driver interconnection line **430** and between the output pad and the panel interconnection line **530** to form an electrical conduction path between the opposite electrodes, while the adhesive may be filled between the adjacent electrodes to insulate the adjacent electrodes from each other.

(26) Hereinafter, the COF package **100** according to the present embodiment will be described in detail.

(27) FIGS. 2A to 2E are schematic diagrams of a COF package **100** according to an embodiment.

(28) Specifically, FIG. 2A is a plan view of the COF package **100**, and FIG. 2B is a bottom view of the COF package **100**. Also, FIG. 2C is a cross-sectional view taken along line C-C' of FIG. 2A, FIG. 2D is a cross-sectional view taken along line D-D' of FIG. 2A, and FIG. 2E is a cross-sectional view taken along line E-E' of FIG. 2A.

(29) Referring to FIGS. 2A to 2E, the COF package **100** may include a base film **110**, a conductive interconnection **120** and a conductive pad CP, which are formed in the base film **110**, and a semiconductor chip SC mounted on the base film **110**.

(30) The base film **110** may be a flexible film including polyimide, which is a material having a high coefficient of thermal expansion and a high durability. A material for the base film **110** is not limited thereto and may include, for example, synthetic resins, such as an epoxy resin, acryl, polyether nitrile, polyether sulfone, polyethylene terephthalate, and polyethylene naphthalate.

(31) The base film **110** may include a circuit region **111** confined inside a cutting line **101** and perforation (PF) regions **112** located at both side ends of the base film **110**. The circuit region **111** may be a region in which the semiconductor chip SC is mounted. In some embodiments, the cutting line **101** may be an imaginary boundary/partition line.

(32) The PF regions **112** may be located at both side ends of the base film **110** and include a plurality of PF holes **114**. Reeling of the base film **110** around a winding reel (not shown) and releasing the base film **110** from the winding reel may be controlled by the PF holes **114**.

(33) In general, since the PF holes **114** are arranged at a constant pitch, a length of the base film **110** may be determined by the number of PF holes **114**. Meanwhile, a width and the length of the base film **110** may be determined by the number and size of semiconductor chips SC mounted on the base film **110** and structures of the conductive interconnection **120** and the conductive pad CP, which are formed in the base film **110**.

(34) Although not shown, a protective layer may be formed on a top surface **110T** and a bottom surface **110B** of the base film **110** to protect/prevent the conductive interconnection **120** from being physically and/or chemically damaged due to external factors. The protective layer may cover the conductive interconnection **120** formed on the top surface **110T** and the bottom surface **110B** of the base film **110** while exposing a predetermined portion of the conductive interconnection **120**. The protective layer may include, for example, solder resist or dry film resist. Alternatively, the protective layer may include a typical insulating film including silicon oxide or silicon nitride.

(35) The conductive interconnection **120** may include a first conductive line **121**, a second conductive line **122**, and a conductive via **123**. The conductive interconnection **120** may include, for example, aluminium foil or copper foil. In some embodiments, the conductive interconnection **120** may be formed by patterning a metal layer formed on the base film **110** using, for example, a

casting process, a laminating process, or an electroplating process.

(36) The conductive interconnection **120** may include the first conductive line **121**, which is formed on the top surface **110T** of the base film **110**, and the second conductive line **122**, which is formed on the bottom surface **110B** of the base film **110**. That is, the conductive interconnection **120** may be formed on both the top surface **110T** and the bottom surface **110B** of the base film **110**. Also, the conductive interconnection **120** may include the conductive via **123**, which passes through the base film **110** and electrically connects the first conductive line **121** to the second conductive line **122**. In some embodiments, a portion of the first conductive line **121** (e.g., a portion adjacent a bonding pad **131** or **132**) on the top surface **110T** of the base film **110** may overlap, in the Z direction, a portion of the second conductive line **122** on the bottom surface **110B** of the base film **110**.

(37) The conductive pad CP may be formed at each of portions adjacent to edges of one end and another end of the base film **110** and connected to the conductive interconnection **120**. The conductive pad CP may be formed on the top surface **110T** of the base film **110**.

(38) The conductive pad CP may include a first row bonding pad **131** and a second row bonding pad **132**, which are connected to a display panel (refer to **500** in FIG. 1), driver PCB **400**, a driver connection pad **150**, which is connected to a driver PCB (refer to **400** in FIG. 1), and a test pad **140** configured to test the conductive interconnection **120**. As used herein, the terms “first row bonding pad” and “second row bonding pad” may refer to first and second rows, respectively, of bonding pads.

(39) The conductive pad CP may be a first portion of the conductive interconnection **120** or a second portion of the conductive interconnection **120** that is plated with tin (Sb), gold (Au), nickel (Ni), or lead (Pb). In some embodiments, the conductive pad CP may be electrically connected to the conductive interconnection **120** and include a conductive material that is separately formed.

(40) The semiconductor chip SC may be a DDI used to drive a display. For example, the semiconductor chip SC may be a source driver chip **300** configured to generate an image signal using a data signal transmitted from a timing controller and output the image signal to the display panel (refer to **500** in FIG. 1). Alternatively, the semiconductor chip SC may be a gate driver chip **200** configured to output a scan signal including an on/off signal of a transistor to the display panel (refer to **500** in FIG. 1).

(41) Although one gate driver chip **200** and two source driver chips **300** are illustrated, types and the number of the semiconductor chips SC are not limited thereto. However, the number of source driver chips **300** may be greater than or equal to the number of gate driver chips **200** in view of characteristics of a display apparatus (refer to **1000** in FIG. 1).

(42) The semiconductor chip SC is not limited to the gate driver chip **200** or the source driver chip **300**. For example, when the COF package **100** is combined with an electronic device other than the display apparatus **1000**, the semiconductor chip SC may be a chip configured to drive the corresponding electronic device.

(43) The semiconductor chip SC may be disposed in the circuit region **111** of the base film **110** and mounted on the base film **110** using a flip-chip bonding process. In other words, a connection terminal **221** (FIG. 2D), such as a solder ball, may be arranged on a chip pad **211** (FIG. 2D) that is exposed at an active surface of the gate driver chip **200**, and a connection terminal **321** (FIG. 2C), such as a solder ball, may be arranged on a chip pad **311** (FIG. 2C) that is exposed at an active surface of each of a plurality of source driver chips **300**. Also, the connection terminals **221** and **321** may be physically and electrically combined with the conductive interconnection **120**, and thus, the gate driver chip **200** and the plurality of source driver chips **300** may be mounted on the base film **110**.

(44) A portion of the chip pad **211** of the gate driver chip **200** and a portion of the chip pad **311** of each of the plurality of source driver chips **300** may serve as input terminals, and another portion of the chip pad **211** of the gate driver chip **200** and another portion of the chip pad **311** of each of the

plurality of source driver chips **300** may serve as output terminals.

(45) Although not shown, the semiconductor chip SC may be sealed with a sealing member to protect/prevent the semiconductor chip SC from being physically and/or chemically damaged due to external factors. Furthermore, an underfill (not shown) may be filled between the semiconductor chip SC and the base film **110**. The underfill may be formed using, for example, a capillary underfill process. The underfill may include, for example, an epoxy resin, but the inventive concept is not limited thereto.

(46) In recent years, as the downscaling of a bezel and the thinning of a panel have been increasingly demanded in the display apparatus **1000**, types and the number of semiconductor chips SC mounted in one COF package **100** are showing a tendency to gradually increase. Thus, the numbers of conductive interconnections **120** and conductive pads CP, which are connected to the semiconductor chips SC, may also increase. Accordingly, the efficient arrangement of the conductive interconnections **120** and the conductive pads CP at a predetermined bonding pitch in a limited circuit region **111** of the base film **110** may significantly affect improvements in the productivity and economical efficiency of the COF package **100**.

(47) As used herein, a bonding pitch may refer to a distance measured between two corresponding points of neighboring conductive interconnections **120** in a direction perpendicular to a lengthwise direction of the conductive interconnections **120**, or a distance measured between two corresponding central points of neighboring conductive pads CP. For instance, the bonding pitch may refer to a distance between centers of two neighboring first conductive lines **121**, which is measured in an X direction, or a distance between centers of two neighboring first row bonding pads **131**, which is measured in the X direction.

(48) Accordingly, the COF package **100** according to the present embodiment may propose a new design for implementing the conductive interconnection **120** and the conductive pad CP, which have a predetermined bonding pitch, while arranging different types of semiconductor chips SC, that is, at least one gate driver chip **200** and at least two source driver chips **300**, on the base film **110**.

(49) To this end, the conductive pad CP connected to the display panel (refer to **500** in FIG. **1**) may include the first row bonding pad **131** and the second row bonding pad **132**. Here, the conductive interconnection **120** may be configured such that the first row bonding pad **131** is connected (e.g., electrically connected) only to the source driver chip **300**, and the conductive interconnection **120** may be configured such that the second row bonding pad **132** is connected (e.g., electrically connected) to both the gate driver chip **200** and the source driver chip **300**. For example, the second row of bonding pads **132** may include a first bonding pad **132** that is electrically connected to the gate driver chip **200** through a first conductive via **123**, and may further include a second bonding pad **132** that is electrically connected to the source driver chip **300** through a second conductive via **123**.

(50) A distance between the first row bonding pad **131** and the source driver chip **300** may be less than a distance between the second row bonding pad **132** and the source driver chip **300**, and the first row bonding pad **131** and the second row bonding pad **132** may be arranged in zigzag in X and Y directions. Accordingly, the first row bonding pad **131** and the second row bonding pad **132** may not be collinear in the Y direction in which they are spaced apart from the source driver chip **300**.

(51) The source driver chip **300** may include a first source driver chip **301** and a second source driver chip **302**. The first source driver chip **301** may have some signal connections to the first row bonding pad **131** and have some other signal connections to the second row bonding pad **132**. Similarly, the second source driver chip **302** may have some signal connections to the first row bonding pad **131** and have some other signal connections to the second row bonding pad **132**.

(52) A connection relationship between the source driver chip **300** and the first and second row bonding pads **131** and **132** will now be described in detail. To begin with, the first row bonding pad **131** may be connected (e.g., electrically connected) to the source driver chip **300** through the first

conductive line **121** (i.e., through the top surface **110T** of the base film **110**) without passing through the conductive via **123**. Next, the second row bonding pad **132** may be indirectly connected (e.g., electrically connected) to the source driver chip **300** through a first conductive via **123**, which is arranged between the first row bonding pad **131** and the source driver chip **300**, and a second conductive via **123**, which is arranged between the second row bonding pad **132** and the cutting line **101** that is close to the second row bonding pad **132**.

(53) Unlike the source driver chip **300**, the gate driver chip **200** may have all signal connections to the second row bonding pad **132**.

(54) A connection relationship between the gate driver chip **200** and the second row bonding pad **132** will now be described in detail. The second row bonding pad **132** may be indirectly connected (e.g., electrically connected) to the gate driver chip **200** through a first conductive via **123**, which is arranged between the gate driver chip **200** and the source driver chip **300**, and a second conductive via **123**, which is arranged between the second row bonding pad **132** and the cutting line **101** that is close to the second row bonding pad **132**.

(55) That is, in view of the configuration of the semiconductor chip SC, the source driver chips **300**, which may use a relatively large number of conductive pads CP for signal connection and are provided in a relatively large number, may exclusively use the first row bonding pad **131**, and additional conductive pads CP may use a portion of the second row bonding pad **132**. In this case, since the first conductive line **121** configured to connect the source driver chip **300** to the first row bonding pad **131** is formed on the top surface **110T** of the base film **110**, the first conductive line **121** may be relatively free in an arrangement relationship with the second conductive line **122** formed on the bottom surface **110B** of the base film **110**. In addition, since the source driver chip **300** is located a relatively small distance from the first row bonding pad **131** and the first conductive line **121** is formed in a straight shape, a routing path may be reduced/minimized, thus facilitating fast signal transmission. That is, the conductive interconnection **120** may be designed to have a predetermined bonding pitch on the top surface **110T** of the base film **110**.

(56) Furthermore, in view of the configuration of the semiconductor chip SC, the gate driver chips **200**, which use a relatively small number of conductive pads CP for signal connection and are provided in a relatively small number, may use the remaining portion of the second row bonding pad **132**. In this case, since the second conductive line **122** configured to connect the gate driver chip **200** to the second row bonding pad **132** is formed on the bottom surface **110B** of the base film **110**, the second conductive line **122** may be relatively free in an arrangement relationship with the first conductive line **121** formed on the top surface **110T** of the base film **110**. Although a portion of the second row bonding pad **132** is electrically connected to the source driver chip **300**, may signal connections of the source driver chip **300** may use the first row bonding pad **131**. That is, the conductive interconnection **120** may be designed to have a predetermined bonding pitch on the bottom surface **110B** of the base film **110**.

(57) Unlike a typical memory semiconductor package, the COF package **100** may include the test pad **140** formed in the base film **110**. The test pad **140** may be used to test a connection failure, such as short circuits in the first conductive line **121**, the second conductive line **122**, and the conductive via **123**. However, since the test pad **140** may play an immaterial role after checking a test result and determining whether the COF package **100** has a connection failure, the test pad **140** may be arranged outside the cutting line **101** (i.e., outside the circuit region **111**).

(58) The test pad **140** configured to test the conductive interconnection **120**, which connects the source driver chip **300** to the first row bonding pad **131**, may be arranged on the top surface **110T** of the base film **110**. From among the conductive vias **123**, a conductive via **123** arranged between the second row bonding pad **132** and the test pad **140** may be arranged outside the circuit region **111**. That is, the COF package **100** may be designed to reduce/minimize the number of conductive vias **123** arranged in the circuit region **111** such that the conductive interconnection **120** has a predetermined bonding pitch.

(59) In addition, the test pad **140** configured to test the conductive interconnection **120**, which connects the gate driver chip **200** to the second row bonding pad **132**, may be arranged on the top surface **110T** of the base film **110**.

(60) In some embodiments, the test pad **140** may be arranged only in a region adjacent to the second row bonding pad **132**. In other words, the test pad **140** may be arranged only in a region adjacent to one edge of the base film **110**. However, the arrangement of the test pad **140** is not limited thereto.

(61) As a result, in the COF package **100** according to the present embodiment, while arranging at least one gate driver chip **200** and at least two source driver chips **300** on the base film **110**, the conductive interconnection **120** and the conductive pad CP, which have a predetermined bonding pitch, may be designed not to intersect or overlap each other, and the test pad **140** configured to test the conductive interconnection **120** may be arranged so as not to affect routing. Thus, the productivity and economical efficiency of the COF package **100** may be improved.

(62) FIG. **3** is a schematic plan view of a portion of a COF package **100a** according to an embodiment.

(63) Most components included in the COF package **100a** that will be described below and materials included in the components are substantially the same as or similar to those described with reference to FIGS. **2A** to **2E**. Thus, differences between the COF package **100a** according to the present embodiment and the COF package (refer to **100** in FIG. **2A**) described above may mainly be described for brevity.

(64) FIG. **3** illustrates first and second row bonding pads **131** and **132** and a test pad **140**, which is electrically connected to the first and second row bonding pads **131** and **132** through a first conductive line **121**, at a top surface **110T** of a base film **110**.

(65) The test pad **140** may include a first row test pad **141** and a second row test pad **142**. A distance between the first row test pad **141** and the second row bonding pad **132** may be less than a distance between the second row test pad **142** and the second row bonding pad **132**, and the first row test pad **141** and the second row test pad **142** may be arranged in zigzag in X and Y directions.

(66) Since the first and second row test pads **141** and **142** may play immaterial roles after checking a test result and determining a connection failure in the COF package **100a**, the first and second row test pads **141** and **142** may be arranged outside a cutting line **101** (i.e., outside a circuit region **111**).

(67) The first row bonding pad **131** and the second row bonding pad **132** may be arranged in parallel in an X direction (e.g., may be collinear in a Y direction). That is, unlike the first and second row test pads **141** and **142**, the first row bonding pad **131** and the second row bonding pad **132** may not be arranged in zigzag in the X and Y directions.

(68) Thus, the first row bonding pad **131** may be connected to the second row test pad **142** through the first conductive line **121**, and the second row bonding pad **132** may be connected to the first row test pad **141** through the first conductive line **121**.

(69) Also, the first conductive line **121** configured to connect the first row bonding pad **131** to the second row test pad **142** may have at least one bent portion, and the first conductive line **121** configured connect the second row bonding pad **132** to the first row test pad **141** may have a straight shape.

(70) Although not shown, in other embodiments, the first row bonding pad **131** and the second row bonding pad **132** may be arranged in zigzag in the X and Y directions. In this case, the first row test pads **141** and the second row test pads **142** may be arranged in parallel in the X direction (e.g., may be collinear in the Y direction). That is, the first row test pad **141** and the second row test pad **142** may not be arranged in zigzag in the X and Y directions.

(71) FIG. **4** is a schematic plan view of a COF package **100b** according to an embodiment.

(72) Most components included in the COF package **100b** that will be described below and materials included in the components are substantially the same as or similar to those described

with reference to FIGS. 2A to 2E. Thus, differences between the COF package **100b** according to the present embodiment and the COF package (refer to **100** in FIG. 2A) described above may mainly be described for brevity.

(73) Referring to FIG. 4, the COF package **100b** may include a base film **110**, a conductive interconnection **120** and a conductive pad CP, which are formed in the base film **110**, and a semiconductor chip SC mounted on the base film **110**.

(74) Although one gate driver chip **200** and two source driver chips **300** are illustrated, types and the number of the semiconductor chips SC are not limited thereto. However, the number of source driver chips **300** may be greater than or equal to the number of gate driver chips **200** in view of characteristics of a display apparatus (refer to **1000** in FIG. 1).

(75) The gate driver chip **200** may include a first gate driver chip **201** and a second gate driver chip **202**. The first gate driver chip **201** may have a signal connection to the second row bonding pad **132**. Similarly, the second gate driver chip **202** may have a signal connection to the second row bonding pad **132**.

(76) The gate driver chip **200** and the source driver chip **300** may be arranged in parallel in an X direction (e.g., may be collinear in a Y direction). Also, the gate driver chip **200** may be arranged a predetermined distance apart from the source driver chip **300** in the Y direction.

(77) A connection relationship between the first gate driver chip **201** and the second row bonding pad **132** will now be described in detail. The second row bonding pad **132** may be indirectly connected to the first gate driver chip **201** through a first conductive via **123**, which is arranged between the first gate driver chip **201** and a first source driver chip **301**, and a second conductive via **123**, which is arranged between the second row bonding pad **132** and a cutting line **101** that is close to the second row bonding pad **132**.

(78) A connection relationship between the second gate driver chip **202** and the second row bonding pad **132** will now be described in detail. The second row bonding pad **132** may be indirectly connected to the second gate driver chip **202** through a first conductive via **123**, which is arranged between the second gate driver chip **202** and a second source driver chip **302**, and a second conductive via **123**, which is arranged between the second row bonding pad **132** and the cutting line **101** that is close to the second row bonding pad **132**.

(79) The COF package **100b** according to the present embodiment may propose a new design for implementing the conductive interconnection **120** and the conductive pad CP, which have a predetermined bonding pitch, while arranging at least two gate driver chips **200** and at least two source driver chips **300** on the base film **110**.

(80) FIGS. 5A to 5E are schematic diagrams of a COF package **100c** according to an embodiment.

(81) Specifically, FIG. 5A is a plan view of the COF package **100c**, and FIG. 5B is a bottom view of the COF package **100c**. Also, FIG. 5C is a cross-sectional view taken along line C-C' of FIG. 5A, FIG. 5D is a cross-sectional view taken along line D-D' of FIG. 5A, and FIG. 5E is a cross-sectional view taken along line E-E' of FIG. 5A.

(82) Most components included in the COF package **100c** that will be described below and materials included in the components are substantially the same as or similar to those described with reference to FIGS. 2A to 2E. Thus, differences between the COF package **100c** according to the present embodiment and the COF package (refer to **100** in FIG. 2A) described above may mainly be described for brevity.

(83) Referring to FIGS. 5A to 5E, the COF package **100c** may include a base film **110**, a conductive interconnection **120** and a conductive pad CP, which are formed in the base film **110**, and a semiconductor chip SC mounted on the base film **110**.

(84) The conductive interconnection **120** may include a first conductive line **121**, a second conductive line **122**, and a conductive via **123**. The conductive interconnection **120** may include the first conductive line **121**, which is formed on a top surface **110T** of the base film **110**, and the second conductive line **122**, which is formed on a bottom surface **110B** of the base film **110**. That

is, the conductive interconnection **120** may be formed on both the top surface **110T** and the bottom surface **110B** of the base film **110**. Also, the conductive interconnection **120** may include the conductive via **123**, which passes through the base film **110** and electrically connects the first conductive line **121** to the second conductive line **122**.

(85) The conductive pad CP may be formed at each of portions adjacent to edges of one end and another end of the base film **110** and connected to the conductive interconnection **120**. The conductive pad CP may be formed on the top surface **110T** or the bottom surface **110B** of the base film **110**. Specifically, a driver connection pad **150** may be arranged on the top surface **110T** of the base film **110** and connected to a driver PCB (refer to **400** in FIG. **1**). First and second row bonding pads **131** and **132** and first and second row test pads **141** and **142** may be provided on the bottom surface **110B** of the base film **110**. The first and second row bonding pads **131** and **132** may be connected to a display panel (refer to **500** in FIG. **1**), and the first and second row test pads **141** and **142** may be used to test the conductive interconnection **120**.

(86) The COF package **100c** according to the present embodiment may propose a new design for implementing the conductive interconnection **120** and the conductive pad CP, which have a predetermined bonding pitch, while arranging at least one gate driver chip **200** and at least two source driver chips **300** on the base film **110**.

(87) Although one gate driver chip **200** and two source driver chips **300** are illustrated, the inventive concept is not limited thereto. However, the number of source driver chips **300** may be greater than or equal to the number of gate driver chips **200** in view of characteristics of a display apparatus (refer to **1000** in FIG. **1**).

(88) The conductive pad CP connected to a display panel (refer to **500** in FIG. **1**) may include the first row bonding pad **131** and the second row bonding pad **132**. Both the first row bonding pad **131** and the second row bonding pad **132** may be arranged on the bottom surface **110B** of the base film **110**.

(89) Here, the conductive interconnection **120** may be configured such that the first row bonding pad **131** is connected only to the gate driver chip **200**, and the conductive interconnection **120** may be connected such that the second row bonding pad **132** is connected only to the source driver chip **300**.

(90) A distance between the first row bonding pad **131** and the source driver chip **300** may be less than a distance between the second row bonding pad **132** and the source driver chip **300**, and the first row bonding pad **131** and the second row bonding pad **132** may be arranged in zigzag in X and Y directions.

(91) The source driver chip **300** may include a first source driver chip **301** and a second source driver chip **302**. The first source driver chip **301** may have a signal connection only to the second row bonding pad **132**. Similarly, the second source driver chip **302** may have a signal connection only to the second row bonding pad **132**.

(92) A connection relationship between the source driver chip **300** and the second row bonding pad **132** will now be described in detail. The source driver chip **300** may have all signal connections to the second row bonding pad **132**. The second row bonding pad **132** may be connected to the source driver chip **300** through the first conductive line **121** and the conductive via **123**, which is arranged between the second row bonding pad **132** and a cutting line **101** that is close to the second row bonding pad **132**.

(93) Also, a connection relationship between the gate driver chip **200** and the first row bonding pad **131** will now be described in detail. The gate driver chip **200** may have all signal connections to the first row bonding pad **131**. The first row bonding pad **131** may be connected to the gate driver chip **200** through the second conductive line **122** and the conductive via **123**, which is arranged between the gate driver chip **200** and the source driver chip **300**.

(94) That is, in view of the configuration of the semiconductor chip SC, the source driver chips **300**, which use a relatively large number of conductive pads CP for signal connection and are

provided in a relatively large number, may exclusively use the second row bonding pad **132**. In this case, since the first conductive line **121** configured to connect the source driver chip **300** to the second row bonding pad **132** is formed on the top surface **110T** of the base film **110**, the first conductive line **121** may be relatively free in an arrangement relationship with the second conductive line **122** formed on the bottom surface **110B** of the base film **110**.

(95) Furthermore, in view of the configuration of the semiconductor chip SC, the gate driver chips **200**, which may use a relatively small number of conductive pads CP for signal connection and are provided in a relatively small number, may exclusively use the first row bonding pad **131**. In this case, since the second conductive line **122** configured to connect the gate driver chip **200** to the first row bonding pad **131** is formed on the bottom surface **110B** of the base film **110**, the second conductive line **122** may be relatively free in an arrangement relationship with the first conductive line **121** formed on the top surface **110T** of the base film **110**. Here, some of the first row bonding pads **131** may be formed as dummy pads that are not connected to the first conductive line **121** and the second conductive line **122**. Thus, the dummy pads may provide free spaces for arranging the conductive interconnections **120** and the conductive pads CP at a predetermined bonding pitch.

(96) Unlike a typical memory semiconductor package, the COF package **100c** may include a test pad **140** formed in the base film **110**. The test pad **140** may be used to test a connection failure, such as short circuits in the first conductive line **121**, the second conductive line **122**, and the conductive via **123**. However, since the test pad **140** may play an immaterial role after checking a test result and determining a connection failure in the COF package **100c**, the test pad **140** may be arranged outside the cutting line **101** (i.e., outside a circuit region).

(97) The second row test pad **142** configured to test the conductive interconnection **120**, which connects the source driver chip **300** to the second row bonding pad **132**, may be arranged on the bottom surface **110B** of the base film **110**. Similarly, the first row test pad **141** configured to test the conductive interconnection **120**, which connects the gate driver chip **200** to the first row bonding pad **131**, may be arranged on the bottom surface **110B** of the base film **110**.

(98) In some embodiments, the first row test pad **141** connected to the first row bonding pad **131** may be arranged in a region adjacent to the driver connection pad **150**. Also, the second row test pad **142** connected to the second row bonding pad **132** may be arranged in a region adjacent to the second row bonding pad **132**. In other words, the test pad **140** may be divided and arranged in regions adjacent to two opposite edges of the base film **110**. However, the arrangement of the test pad **140** is not limited thereto.

(99) As a result, in the COF package **100c** according to the present embodiment, while arranging at least one gate driver chip **200** and at least two source driver chips **300** on the base film **110**, the conductive interconnection **120** and the conductive pad CP, which have a predetermined bonding pitch, may be designed not to intersect or overlap each other, and the test pad **140** configured to test the conductive interconnection **120** may be arranged so as not to affect routing. Thus, the productivity and economical efficiency of the COF package **100c** may be improved.

(100) FIG. **6** is a schematic plan view of a portion of a COF package **100d** according to an embodiment.

(101) Most components included in the COF package **100d** that will be described below and materials included in the components are substantially the same as or similar to those described with reference to FIGS. **5A** to **5E**. Thus, differences between the COF package **100d** according to the present embodiment and the COF package (refer to **100c** in FIG. **5A**) described above may mainly be described for brevity.

(102) FIG. **6** illustrates a second row test pad **142**, which is electrically connected to a second row bonding pad **132** through a second conductive line **122** at a bottom surface **110B** of a base film **110**.

(103) A test pad (refer to **140** in FIG. **5A**) may include a first row test pad (refer to **141** in FIG. **5A**) and a second row test pad **142**. A distance between the second row test pad **142** and the second row bonding pad **132** may be less than a distance between the second row test pad **142** and a first row

bonding pad **131**, and the first row test pad **141** and the second row test pad **142** may be separately arranged in different regions of the base film **110**.

(104) Since the second row test pad **142** may play an immaterial role after checking a test result and determining whether the COF package **100d** has a connection failure, the second row test pad **142** may be arranged outside a cutting line **101** (i.e., outside a circuit region).

(105) The first row bonding pad **131** and the second row bonding pad **132** may be arranged in parallel in an X direction (e.g., may be collinear in a Y direction). That is, the first row bonding pad **131** and the second row bonding pad **132** may not be arranged in zigzag in X and Y directions.

(106) The first row bonding pad **131** may be electrically connected to the first row test pad **141** by one second conductive line **122**, and the second row bonding pad **132** may be electrically connected to the second row test pad **142** by another second conductive line **122**. Also, each of the second conductive line **122** configured to connect the first row bonding pad **131** to the first row test pad **141** and the second conductive line **122** configured to connect the second row bonding pad **132** to the second row test pad **142** may have a straight shape.

(107) Although not shown, in other embodiments, the first row bonding pad **131** and the second row bonding pad **132** may be arranged in zigzag in X and Y directions.

(108) FIG. 7 is a schematic plan view of a COF package **100e** according to an embodiment.

(109) Most components included in the COF package **100e** that will be described below and materials included in the components are substantially the same as or similar to those described with reference to FIGS. 5A to 5E. Thus, differences between the COF package **100e** according to the present embodiment and the COF package (refer to **100c** in FIG. 5A) described above may mainly be described for brevity.

(110) FIG. 7 illustrates a test pad **140**, which is electrically connected to first and second row bonding pads **131** and **132** through a second conductive line **122** and formed on a bottom surface **110B** of a base film **110**.

(111) Unlike a typical memory semiconductor package, the COF package **100e** may include the test pad **140** in the base film **110**. The test pad **140** may be used to test a connection failure, such as short circuits in the first conductive line **121**, the second conductive line **122**, and the conductive via **123**. However, since the test pad **140** may play an immaterial role after checking a test result and determining whether the COF package **100e** has a connection failure, the test pad **140** may be arranged outside a cutting line **101** (i.e., outside a circuit region).

(112) A test pad **140** configured to test a conductive interconnection **120**, which connects a source driver chip **300** to the second row bonding pad **132**, may be arranged on the bottom surface **110B** of the base film **110**. Similarly, a test pad **140** configured to test a conductive interconnection **120**, which connects a gate driver chip **200** to the first row bonding pad **131**, may be arranged on the bottom surface **110B** of the base film **110**.

(113) In some embodiments, the test pad **140** may be arranged only in a region adjacent to the second row bonding pad **132**. In other words, the test pad **140** configured to test the first and second row bonding pads **131** and **132** may not be divided but arranged only in a region adjacent to one edge of the base film **110**. However, the arrangement of the test pad **140** is not limited thereto.

(114) FIG. 8 is a schematic plan view of a COF package **100f** according to an embodiment.

(115) Most components included in the COF package **100f** that will be described below and materials included in the components are substantially the same as or similar to those described with reference to FIGS. 5A to 5E. Thus, differences between the COF package **100f** according to the present embodiment and the COF package (refer to **100c** in FIG. 5A) described above may mainly be described for brevity.

(116) Referring to FIG. 8, the COF package **100f** may include a base film **110**, a conductive interconnection **120** and a conductive pad CP, which are formed in the base film **110**, and a semiconductor chip SC mounted on the base film **110**.

(117) Although two gate driver chips **200** and two source driver chips **300** are illustrated in FIG. 8,

the inventive concept is not limited thereto. However, the number of source driver chips **300** may be greater than or equal to the number of gate driver chips **200** in view of characteristics of a display apparatus (refer to **1000** in FIG. 1).

(118) The gate driver chip **200** may include a first gate driver chip **201** and a second gate driver chip **202**. The first gate driver chip **201** may have a signal connection only to a first row bonding pad **131**. Similarly, the second gate driver chip **202** may have a signal connection only to the first row bonding pad **131**.

(119) The gate driver chip **200** and the source driver chip **300** may be arranged in parallel in an X direction (e.g., may be collinear in a Y direction). Also, the gate driver chip **200** may be arranged a predetermined distance apart from the source driver chip **300** in the Y direction.

(120) A connection relationship between the first gate driver chip **201** and the first row bonding pad **131** will now be described in detail. The first row bonding pad **131** may be connected to the first gate driver chip **201** through a second conductive line **122** and a conductive via **123**, which is arranged between the first gate driver chip **201** and a first source driver chip **301**.

(121) A connection relationship between the second gate driver chip **202** and a second row bonding pad **132** will now be described in detail. The first row bonding pad **131** may be connected to the second gate driver chip **202** through the second conductive line **122** and the conductive via **123**, which is arranged between the second gate driver chip **202** and a second source driver chip **302**.

(122) The COF package **100f** according to the present embodiment may propose a new design for implementing the conductive interconnection **120** and the conductive pad CP, which have a predetermined bonding pitch, while arranging at least two gate driver chips **200** and at least two source driver chips **300** on the base film **110**.

(123) FIGS. 9A and 9B are schematic diagrams of a display apparatus **1000a** including a COF package **100'** according to an embodiment.

(124) Specifically, FIG. 9A is a cross-sectional view of the display apparatus **1000a**, and FIG. 9B is a partial enlarged view of a region BB of FIG. 9A.

(125) Referring to FIGS. 9A and 9B, the display apparatus **1000a** may include the COF package **100'**, a driver PCB **400**, and a display panel **500**.

(126) The display panel **500** may include a transparent substrate **510** and an image region **520** and a plurality of panel interconnection lines **530**, which are formed on the transparent substrate **510**. In some embodiments, the image region **520** and the plurality of panel interconnection lines **530** may be formed on a front surface **510T** of the transparent substrate **510**. A plurality pixels included in the image region **520** may be connected to the plurality of panel interconnection lines **530**.

(127) Here, the COF package **100'** may refer to only a portion of a circuit region **111**, which is cut along a cutting line **101** in the COF package **100** described above with reference to FIGS. 2A to 2E.

(128) A portion of the COF package **100'** may face a portion of the display panel **500**. Specifically, a portion of a base film **110** on which a bonding pad **130** is arranged may be opposite to a portion of the front surface **510T** of the transparent substrate **510**. The bonding pad **130** may include a first row bonding pad (refer to **131** in FIG. 2A) and a second row bonding pad (refer to **132** in FIG. 2A).

(129) The COF package **100'** may receive a signal output by the driver PCB **400** and transmit the signal to the display panel **500**. At least one driver circuit chip (refer to **410** in FIG. 1) capable of simultaneously or sequentially applying power and a signal to the COF package **100'** may be mounted on the driver PCB **400**.

(130) The COF package **100'** may include a source driver chip **300**, a gate driver chip **200**, a first conductive line (refer to **121** in FIG. 2C) formed on a top surface of the base film **110**, a second conductive line (refer to **122** in FIG. 2C) formed on a bottom surface of the base film **110**, and a conductive via (refer to **123** in FIG. 2C) configured to connect the first and second conductive lines **121** and **122** to each other.

(131) The display panel **500** may have the front surface **510T** of the transparent substrate **510** on which the image region **520** including the plurality of pixels is arranged and a rear surface **510B**

that is opposite to the front surface **510T**. The COF package **100'** may be bent toward the rear surface **510B** of the display panel **500** and fixed (i.e., attached) to the display panel **500**, and the driver PCB **400** may face the rear surface **510B** of the display panel **500**.

(132) The display apparatus **1000a** according to the present embodiment may implement a conductive interconnection (refer to **120** in FIG. 2C) and a conductive pad (refer to CP in FIG. 2A), which have a predetermined bonding pitch, while arranging at least one gate driver chip **200** and at least two source driver chips **300** of the base film **110** of the COF package **100'**. Thus, the productivity and economical efficiency of the display apparatus **1000a** including the COF package **100'** may be improved.

(133) FIGS. **10A** and **10B** are schematic diagrams of a display apparatus **1000b** including a COF package **100c'** according to an embodiment.

(134) Specifically, FIG. **10A** is a cross-sectional view of the display apparatus **1000b**, and FIG. **10B** is a partial enlarged view of a region BB of FIG. **10A**.

(135) Referring to FIGS. **10A** and **10B**, the display apparatus **1000b** may include the COF package **100c'**, a driver PCB **400**, and a display panel **500**.

(136) The display panel **500** may include a transparent substrate **510** and an image region **520** and a plurality of panel interconnection lines **530** formed on the transparent substrate **510**. In some embodiments, the image region **520** and the plurality of panel interconnection lines **530** may be formed on a front surface **510T** of the transparent substrate **510**. A plurality of pixels included in the image region **520** may be connected to the plurality of panel interconnection lines **530**.

(137) Here, the COF package **100c'** may refer to only a portion of a circuit region **111**, which is cut along a cutting line **101** in the COF package **100c** described above with reference to FIGS. 5A to 5E.

(138) A portion of the COF package **100c'** may face a portion of the display panel **500**. Specifically, a portion of a base film **110** on which a bonding pad **130** is arranged may be opposite to a portion of the front surface **510T** of the transparent substrate **510**. The bonding pad **130** may include a first row bonding pad (refer to **131** in FIG. 5A) and a second row bonding pad (refer to **132** in FIG. 5A).

(139) The COF package **100c'** may receive a signal output by the driver PCB **400** and transmit the signal to the display panel **500**. At least one driver circuit chip (refer to **410** in FIG. 1) capable of simultaneously or sequentially applying power and a signal to the COF package **100c'** may be mounted on the driver PCB **400**.

(140) The COF package **100c'** may include a source driver chip **300**, a gate driver chip **200**, a first conductive line (refer to **121** in FIG. 5C) formed on a top surface of the base film **110**, a second conductive line (refer to **122** in FIG. 5C) formed on a bottom surface of the base film **110**, and a conductive via (refer to **123** in FIG. 5C) configured to connect the first and second conductive lines **121** and **122** to each other.

(141) The display panel **500** may have the front surface **510T** of the transparent substrate **510** on which the image region **520** including the plurality of pixels is arranged and a rear surface **510B** that is opposite to the front surface **510T**. The COF package **100c'** may be bent toward the rear surface **510B** of the display panel **500** and fixed to the display panel **500**, and the driver PCB **400** may be arranged apart from the rear surface **510B** of the display panel **500** with the COF package **100c'** therebetween.

(142) The display apparatus **1000b** according to the present embodiment may implement a conductive interconnection (refer to **120** in FIG. 5C) and a conductive pad (refer to CP in FIG. 5A), which have a predetermined pitch, while arranging at least one gate driver chip **200** and at least two source driver chips **300** on the base film **110** of the COF package **100c'**. Thus, the productivity and economical efficiency of the display apparatus **1000b** including the COF package **100c'** may be improved.

(143) While the inventive concept has been particularly shown and described with reference to

example embodiments thereof, it will be understood that various changes in form and details may be made therein without departing from the scope of the following claims.

Claims

1. A display apparatus comprising: a chip-on-film (COF) package having a base film; a display panel that faces a first portion of a top surface of the base film; and a driver printed circuit board (PCB) that faces a second portion of the top surface of the base film, wherein the COF package comprises: the base film; a source driver chip and a gate driver chip each mounted on the top surface of the base film; a first conductive line on the top surface of the base film, a second conductive line on a bottom surface of the base film, and a conductive via that electrically connects the first and second conductive lines to each other; a first row of bonding pads on the top surface of the base film and electrically connected to the source driver chip; and a second row of bonding pads on the top surface of the base film and electrically connected to the source driver chip and the gate driver chip.
2. The display apparatus of claim 1, wherein the base film comprises a flexible film.
3. The display apparatus of claim 2, wherein the display panel has a front surface including a plurality of pixels and a rear surface opposite to the front surface, wherein the COF package is bent toward the rear surface of the display panel and attached to the display panel, and wherein the driver PCB faces the rear surface of the display panel.
4. The display apparatus of claim 1, wherein, in the COF package, the first row of bonding pads is electrically connected to the source driver chip through the first conductive line without passing through the conductive via.
5. The display apparatus of claim 1, wherein, in the COF package, the second row of bonding pads is electrically connected to the source driver chip through the conductive via.
6. The display apparatus of claim 1, wherein, in the COF package, the second row of bonding pads is electrically connected to the gate driver chip through the conductive via.
7. The display apparatus of claim 1, wherein the source driver chip comprises at least two source driver chips and the gate driver chip comprises at least one gate driver chip, and wherein the number of source driver chips is greater than or equal to the number of gate driver chips.
8. The display apparatus of claim 1, wherein, in the COF package, a first distance between the first row of bonding pads and the source driver chip is less than a second distance between the second row of bonding pads and the source driver chip.
9. The display apparatus of claim 8, wherein, in the COF package, the first row of bonding pads and the second row of bonding pads are in a zigzag arrangement.
10. The display apparatus of claim 1, wherein, in the COF package, a portion of the first conductive line on the top surface of the base film overlaps a portion of the second conductive line on the bottom surface of the base film.
11. A display apparatus comprising: a chip-on-film (COF) package having a base film; a display panel that faces a portion of a bottom surface of the base film; and a driver printed circuit board (PCB) that faces a portion of a top surface of the base film, wherein the COF package comprises: the base film; a source driver chip and a gate driver chip each mounted on the top surface of the base film; a first conductive line on the top surface of the base film, a second conductive line on the bottom surface of the base film, and a conductive via that electrically connects the first and second conductive lines to each other; a first row of bonding pads on the bottom surface of the base film and electrically connected to the gate driver chip; and a second row of bonding pads on the bottom surface of the base film and electrically connected to the source driver chip.
12. The display apparatus of claim 11, wherein the base film comprises a flexible film.
13. The display apparatus of claim 12, wherein the display panel has a front surface including a plurality of pixels and a rear surface opposite to the front surface, wherein the COF package is bent

toward the rear surface of the display panel and attached to the display panel, and wherein the COF package is between the driver PCB and the rear surface of the display panel.

14. The display apparatus of claim 11, wherein the source driver chip comprises at least two source driver chips and the gate driver chip comprises at least one gate driver chip, and wherein the number of source driver chips is greater than or equal to the number of gate driver chips.

15. The display apparatus of claim 11, wherein, in the COF package, the first row of bonding pads is electrically connected to the gate driver chip through the conductive via.

16. The display apparatus of claim 11, wherein, in the COF package, the second row of bonding pads is electrically connected to the source driver chip through the conductive via.

17. The display apparatus of claim 11, wherein, in the COF package, a first distance between the first row of bonding pads and the source driver chip is less than a second distance between the second row of bonding pads and the source driver chip.

18. The display apparatus of claim 17, wherein, in the COF package, the first row of bonding pads and the second row of bonding pads are in a zigzag arrangement.

19. The display apparatus of claim 17, wherein, in the COF package, the first row of bonding pads and the second row of bonding pads are in a straight arrangement.

20. The display apparatus of claim 17, wherein, in the COF package, some bonding pads of the first row of bonding pads are dummy pads that are not electrically connected to the first and second conductive lines.
